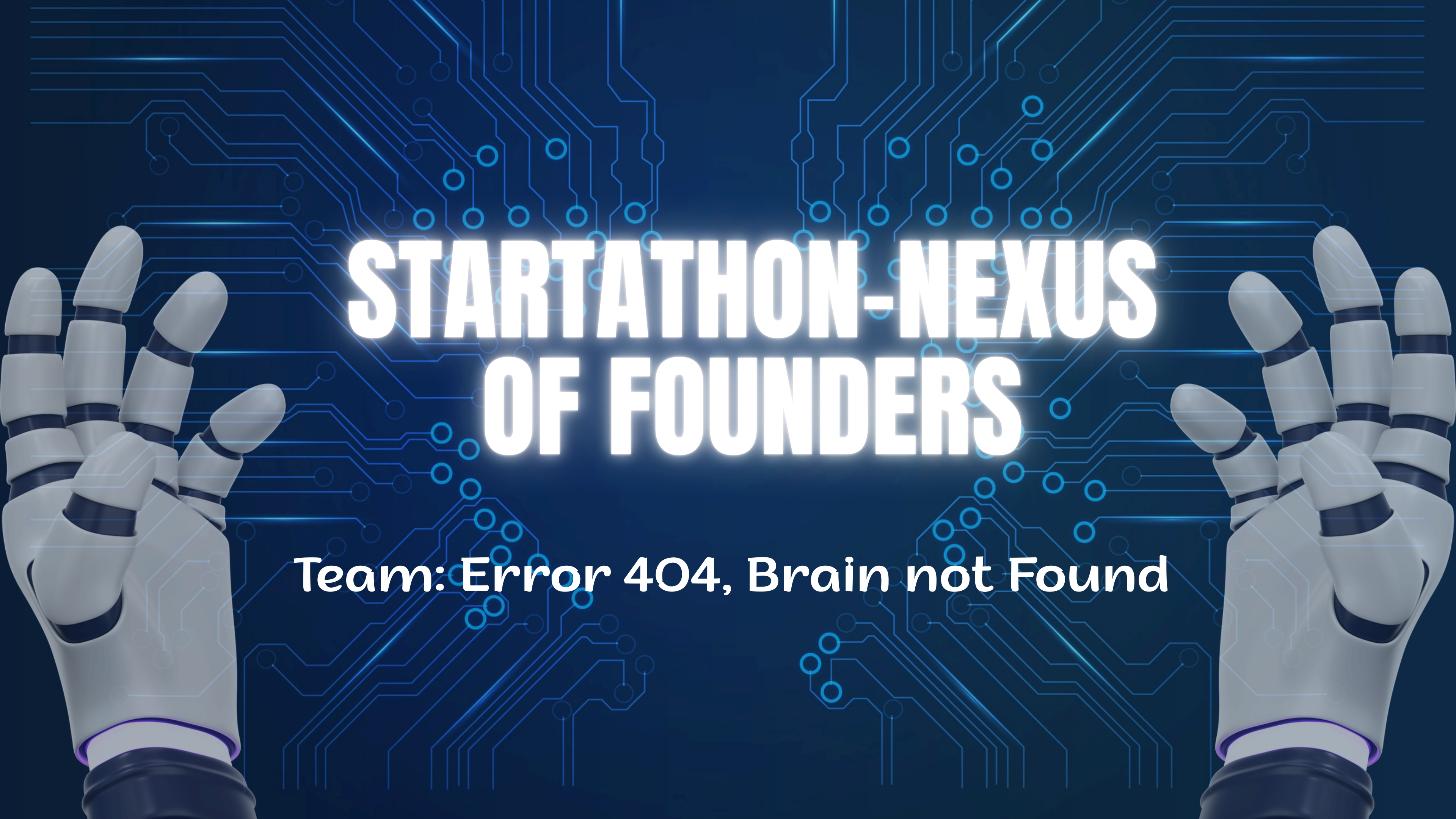




STARTATHON-NEXUS OF FOUNDERS

Team: Error 404, Brain not Found

team work

Ritambhar Advait (25bcy10086)

Raushan kumar (25bsa10037)

Ayushi (25bsa10037)

Harshita singh bais(25bai11599)



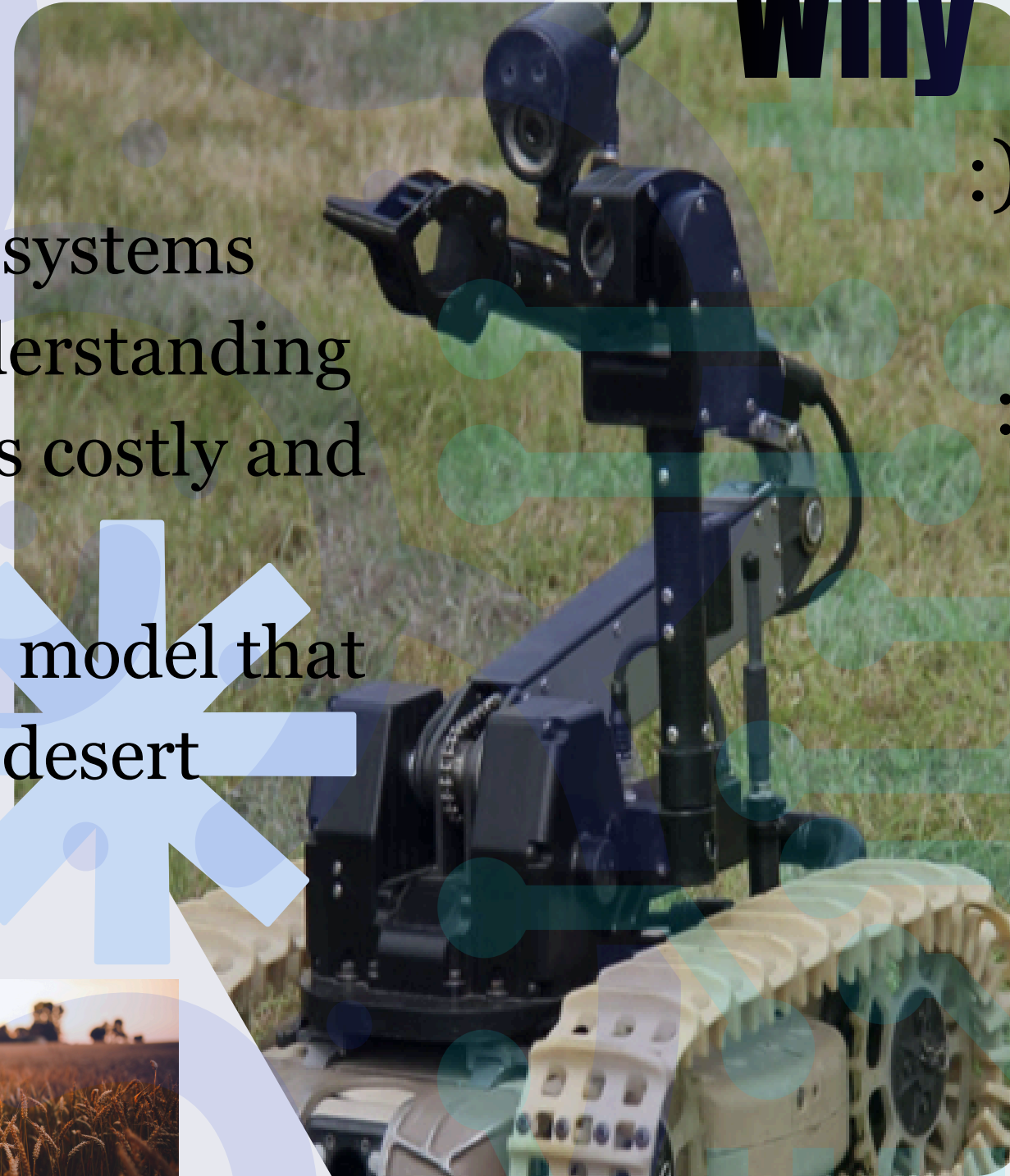
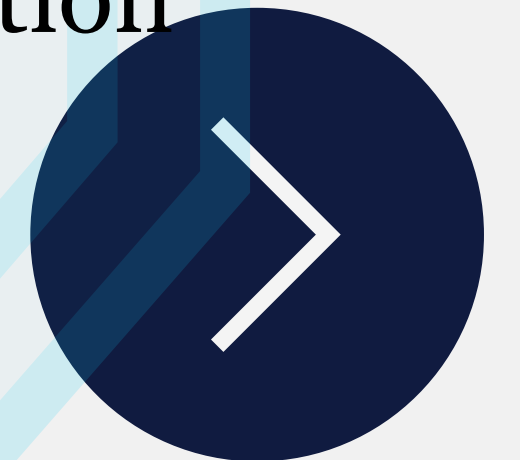
Problem Statement

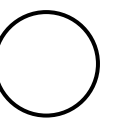
- :)Autonomous off-road systems require accurate scene understanding
- :)Real-World labeled data is costly and limited
- :)Goal: Train segmentation model that generalizes to unseen desert environments



Why it Matters?

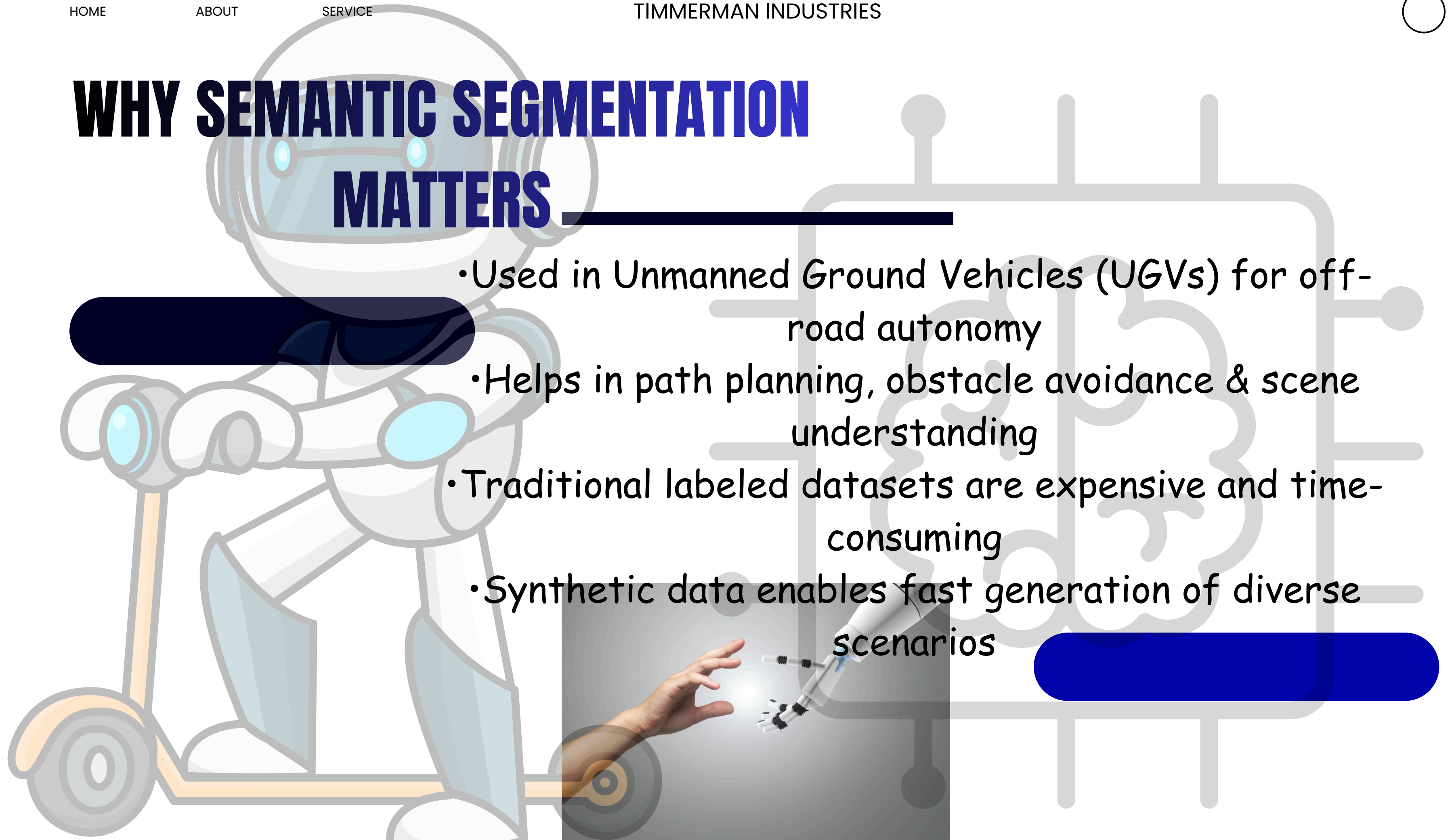
- :)Autonomous ground vehicles
- :)Disaster response robotics
- :)Exploration and agriculture automation
- :)Real-time environmental perception





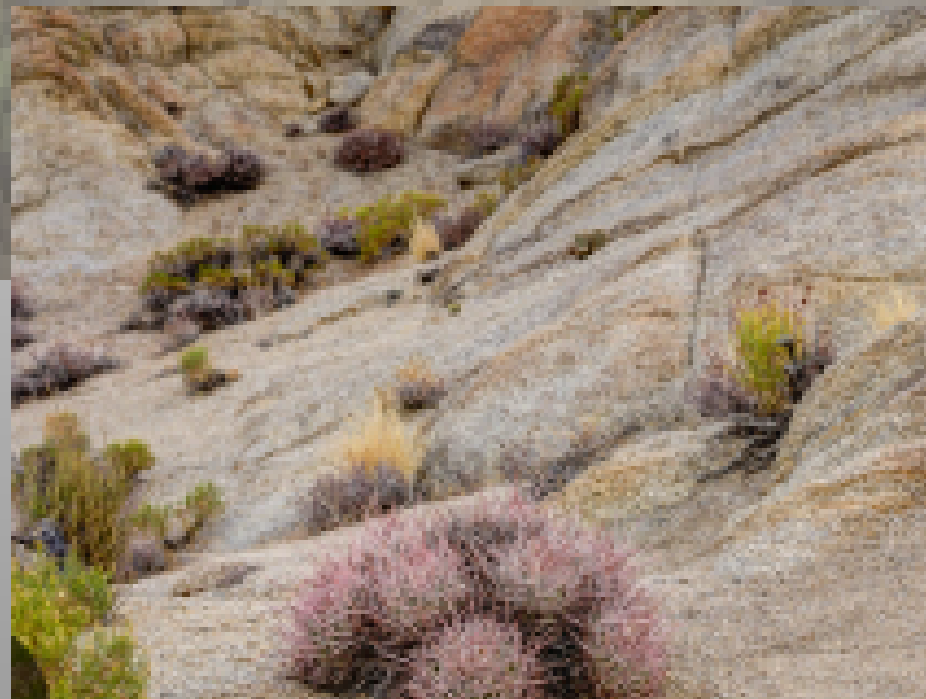
WHY SEMANTIC SEGMENTATION MATTERS

- Used in Unmanned Ground Vehicles (UGVs) for off-road autonomy
- Helps in path planning, obstacle avoidance & scene understanding
- Traditional labeled datasets are expensive and time-consuming
- Synthetic data enables fast generation of diverse scenarios



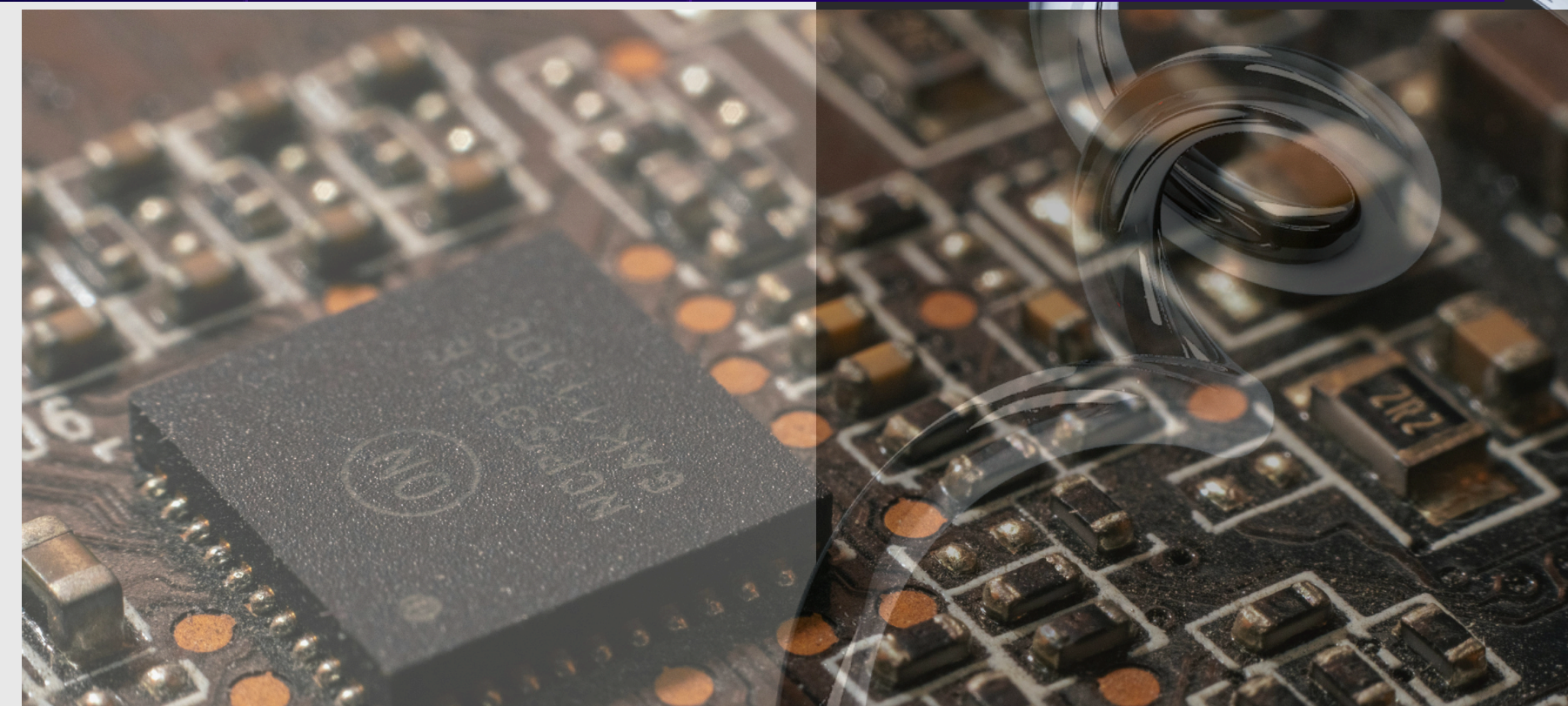
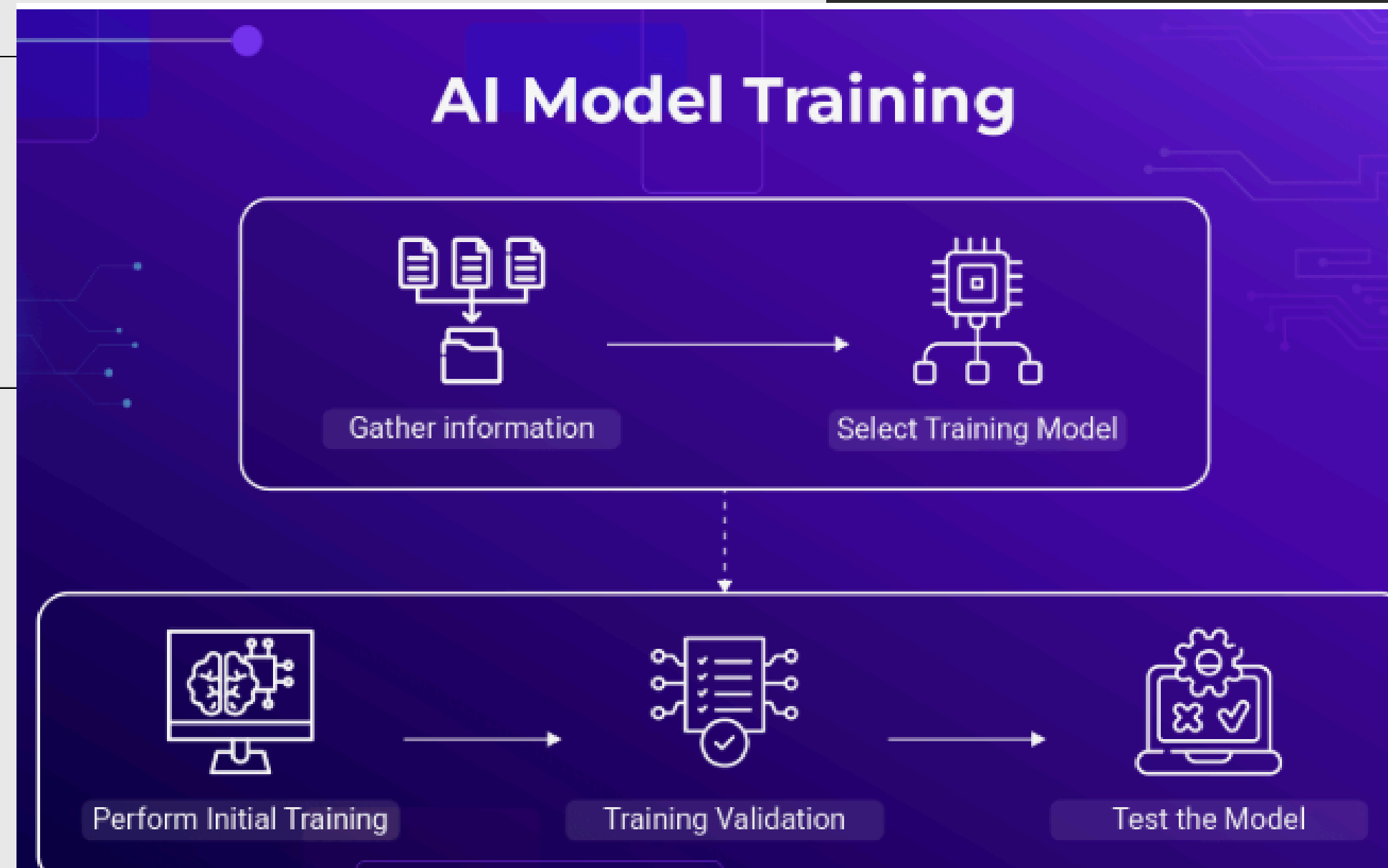
Dataset Overview

Synthetic
desert dataset



Training Workflow

- Setup Falcon account and download dataset
- Setup environment using Miniconda/Anaconda (EDU environment)
- Run training script: `python train.py`
- Check logs & checkpoints inside runs/ directory
- Run test script: `python test.py` (unseen testImages)
- Use IoU score as main benchmark



Training

```

    response = json.loads(_read_url(Request(url, headers=headers)))
File "C:\Users\raush\anaconda3\envs\EDU\lib\site-packages\torch\hub.py", line 185, in _read_url
    with urlopen(url) as r:
File "C:\Users\raush\anaconda3\envs\EDU\lib\urllib\request.py", line 216, in urlopen
    return opener.open(url, data, timeout)
File "C:\Users\raush\anaconda3\envs\EDU\lib\urllib\request.py", line 525, in open
    response = meth(req, response)
File "C:\Users\raush\anaconda3\envs\EDU\lib\urllib\request.py", line 634, in http_response
    response = self.parent.error(
File "C:\Users\raush\anaconda3\envs\EDU\lib\urllib\request.py", line 563, in error
    return self._call_chain(*args)
File "C:\Users\raush\anaconda3\envs\EDU\lib\urllib\request.py", line 496, in _call_chain
    result = func(*args)
File "C:\Users\raush\anaconda3\envs\EDU\lib\urllib\request.py", line 643, in http_error_default
    raise HTTPError(req.full_url, code, msg, hdrs, fp)
urllib.error.HTTPError: HTTP Error 403: rate limit exceeded

```

```
(EDU) C:\Startathon\Offroad_Segmentation_Scripts>conda activate EDU
```

```
(EDU) C:\Startathon\Offroad_Segmentation_Scripts>cd C:\Startathon\
```

```
(EDU) C:\Startathon\Offroad_Segmentation_Scripts>python train_segnet.py
```

```
Using device: cpu
```

Training samples: 2857

Validation samples: 317

Loading DINOv2 backbone...

```
Downloading: "https://github.com/facebookresearch/dinov2/zipball/n
```

Anaconda Prompt - conda ac

```

response = json.loads(_read_url(Request(url, headers=headers)))
File "C:\Users\raush\anaconda3\envs\EDU\lib\site-packages\torch\hub.py", line 185, in _read_url
    with urlopen(url) as r:
File "C:\Users\raush\anaconda3\envs\EDU\lib\urllib\request.py", line 216, in urlopen
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File "C:\Users\raush\anaconda3\envs\EDU\lib\urllib\request.py", line 525, in open
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File "C:\Users\raush\anaconda3\envs\EDU\lib\urllib\request.py", line 643, in http_error_default
    raise HTTPError(req.full_url, code, msg, hdrs, fp)
urllib.error.HTTPError: HTTP Error 403: rate limit exceeded

```

```
(EDU) C:\Startathon\Offroad_Segmentation_Scripts>conda activate EDU
```

```
(EDU) C:\Startathon\Offroad_Segmentation_Scripts>cd C:\Startathon\Offroad_Segmentation_Scripts
```

```
(EDU) C:\Startathon\Offroad_Segmentation_Scripts>python train_segmentation.py
```

Using device: cpu

Training samples: 2857

Validation samples: 317
Leading DTN: 2, backbonesLoading DINOv2 backbone...
Downloading: "https://github.com/

Downloading: <https://github.com/facebookresearch/dinov2/zipball/main> to C:\Users\raushl\AppData\Local\Microsoft\Windows\Temporary Internet Files\Content.IE5\B7Q8Q8Q8\main.zip

```
C:\Users\raush\.cache\torch\nub\facebookresearch_dinov2_main\dinov2\layers\swiglu_+n.py:51: UserWarning: xFormers is not available (SWIGLU)
```

```
warnings.warn("xFormers is not available (SwiGLU)")
```

```
C:\Users\raush/.cache\torch\hub\facebookresearch_dinov2_main\dinov2\layers\attention.py:33: UserWarning: xFormers is not available (Attention)
```

```
warnings.warn("xFormers is not available (Attention)")
```

```
C:\Users\raush/.cache/torch/hub/facebookresearch_dinov2_main\dinov2\layers\block.py:40: UserWarning: xFormers is not available (Block)
```

```
warnings.warn("xFormers is not available (Block)")
```

100% | 8" 3M/8" 3M [02-11-00-00 6711.0/-]

100%	84.2M/84.2M [02:11<00:00, 0.71KB/s]
------	-------------------------------------

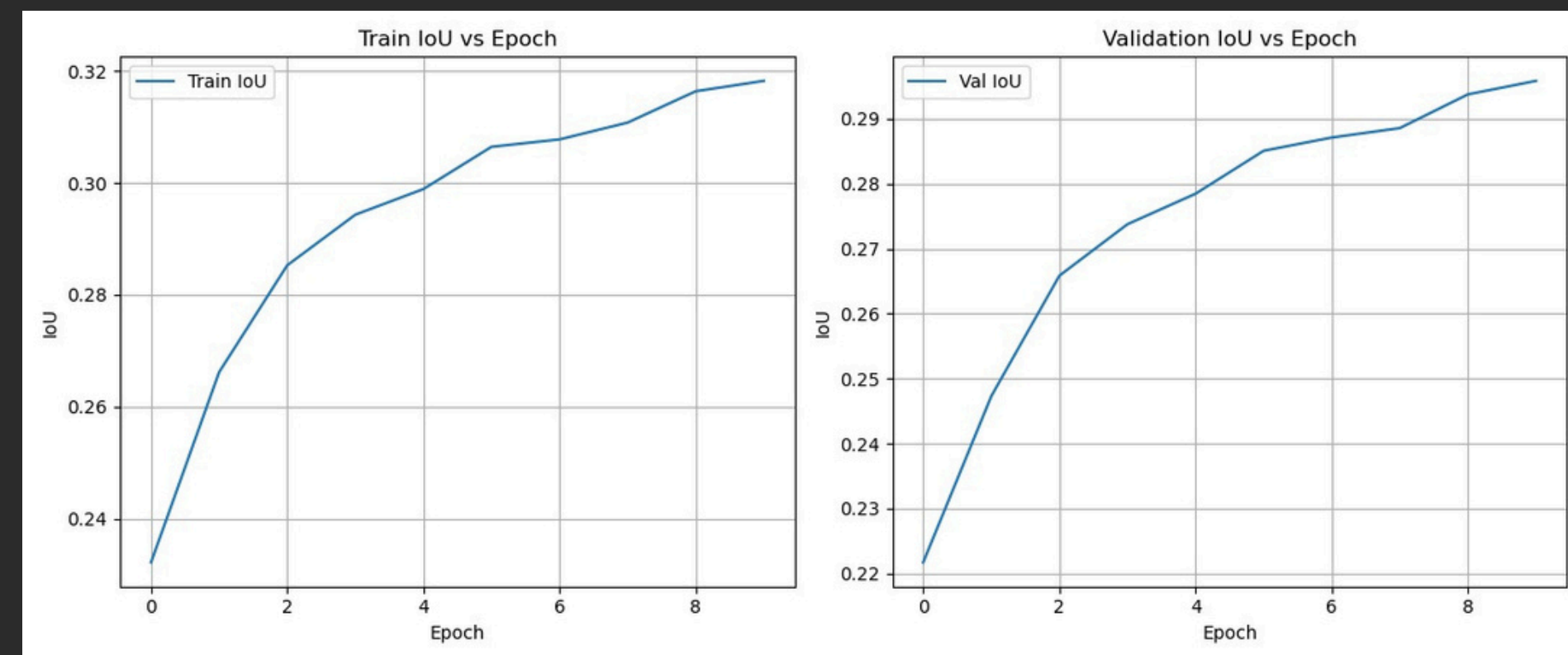
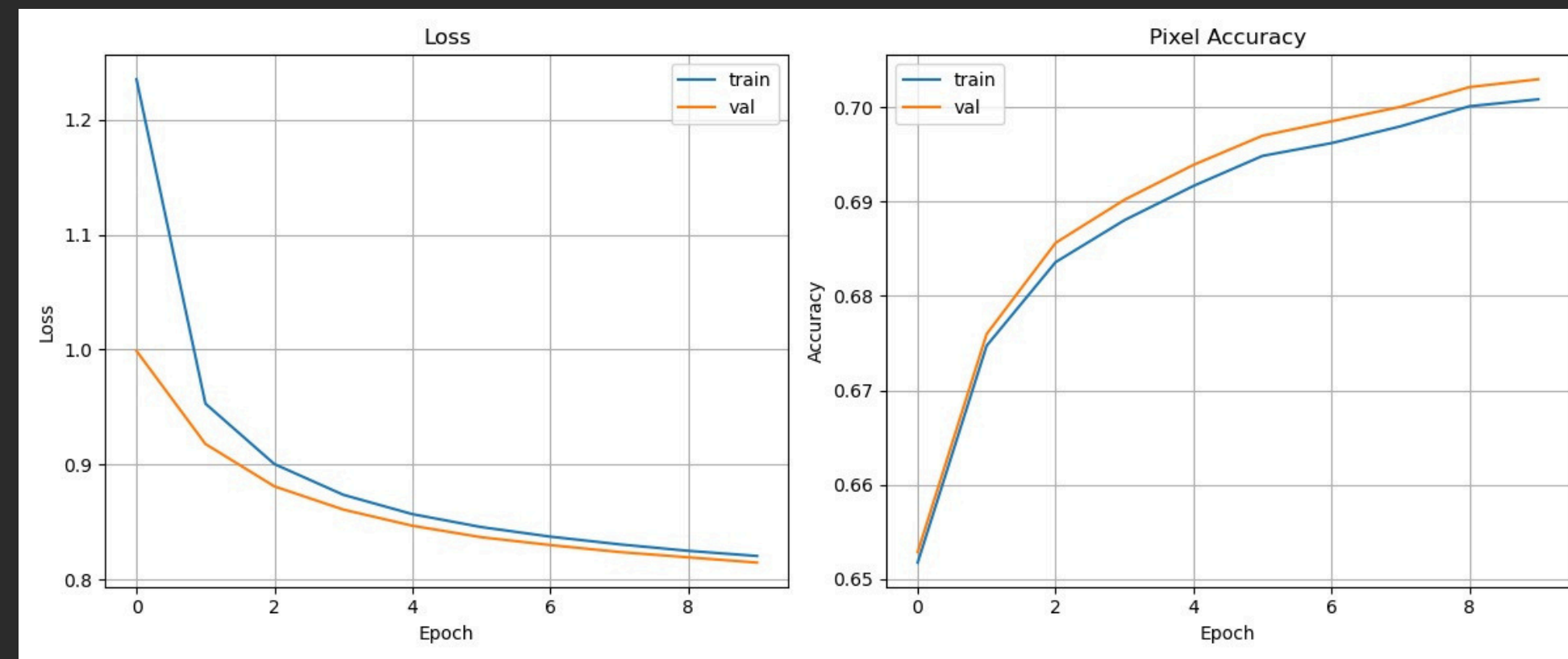
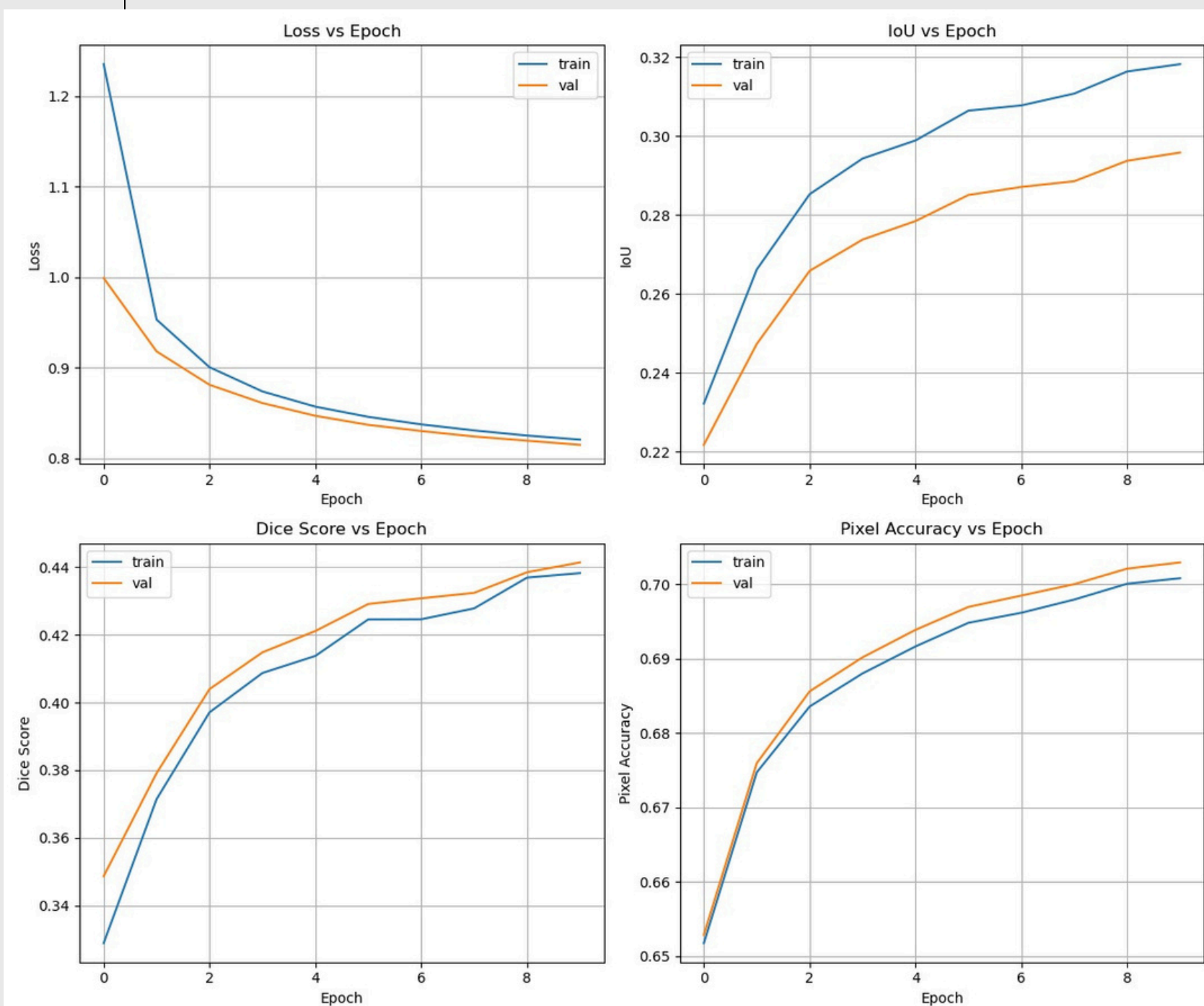
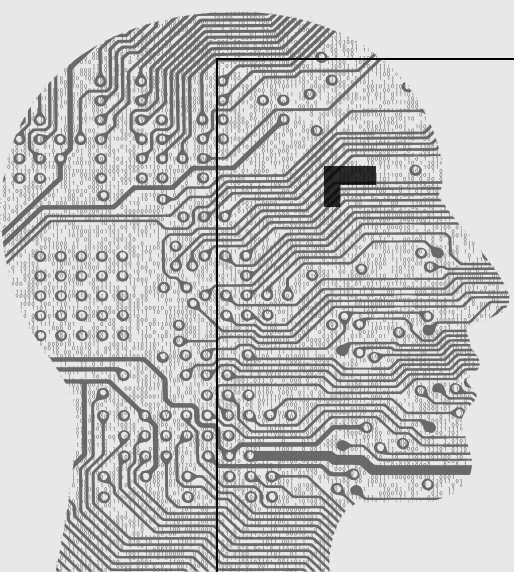
```
Backbone loaded successfully!
```

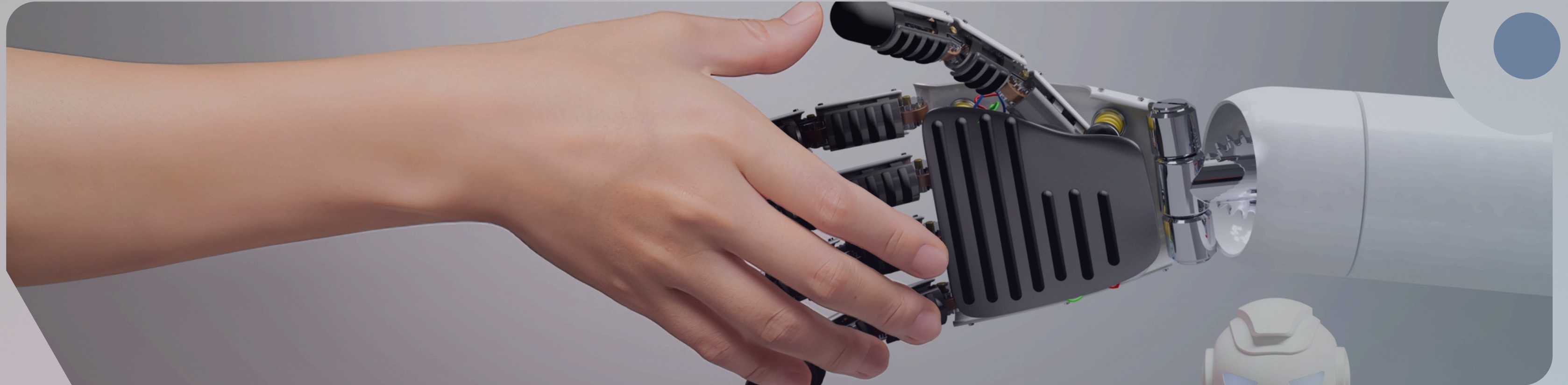
Embedding dimension: 384

```
Patch tokens shape: torch.Size([2, 646, 384])
```

```
Epoch 1/10 [Train]: 8% | ██████████ | 108/1120 [01:04<13:40] 1.61batch/c 1.65571 8.1161
```

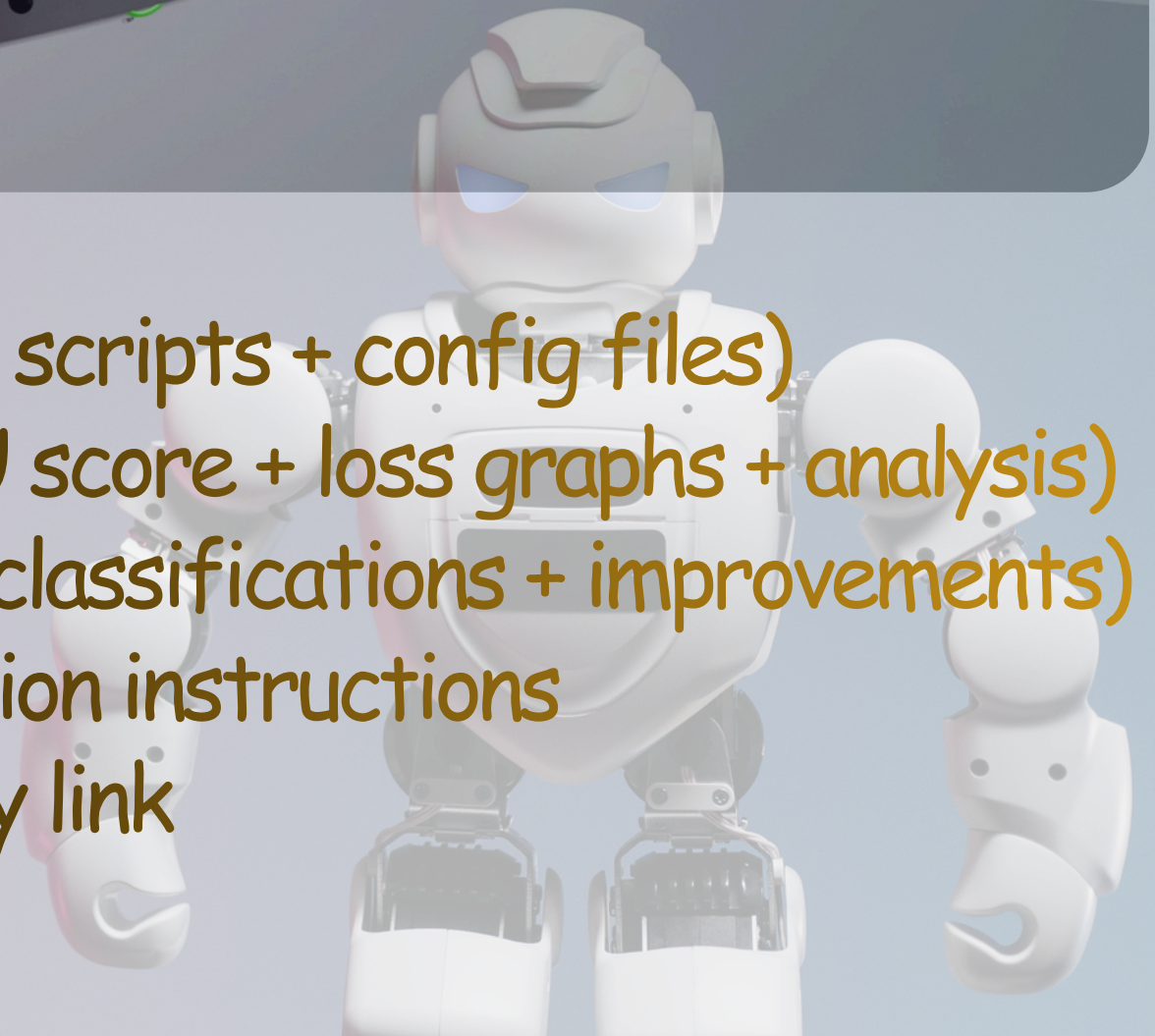
Epoch 1/10 [Train]: 0% | 100/1429 [01.04~13.40, 1.01batch/s, loss=1.0461]





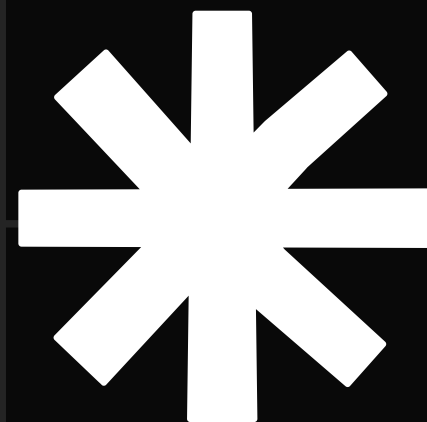
Key Deliverables

- Trained model (weights + scripts + config files)
- Performance report (IoU score + loss graphs + analysis)
- Failure case analysis (misclassifications + improvements)
- README with reproduction instructions
- Private GitHub repository link



...Conclusion *

- Successfully trained a Semantic Segmentation model on Falcon's synthetic offroad desert dataset.
-
- Training was improved using data preprocessing + augmentation + hyperparameter tuning.
-
- Final results include trained model weights, evaluation report, loss/IoU graphs, and visual prediction outputs.
-



Thankyou

